

TCP Socket Programming

server.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(("127.0.0.1", 8080))
server_socket.listen(1)
print("Server is waiting for connection...")

client_socket, client_address = server_socket.accept()
print("Client connected")

message = client_socket.recv(1024).decode()
print("Client:", message)

reply = input("Enter reply: ")
client_socket.send(reply.encode())

client_socket.close()
server_socket.close()
```

client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(("127.0.0.1", 8080))
print("Connected to server")

message = input("Enter message: ")
client_socket.send(message.encode())

reply = client_socket.recv(1024).decode()
print("Server:", reply)

client_socket.close()
```

Server Output

```
$ python3 server.py
Server is waiting for connection...
Client connected
Client: Hello Server
Enter reply: Hello Client
$
```

Client Output

```
$ python3 client.py
Connected to server
Enter message: Hello Server
Server: Hello Client
```

\$

server_math.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(("127.0.0.1", 8080))
server_socket.listen(1)
print("Server is waiting for connection...")

client_socket, client_address = server_socket.accept()
print("Client connected")

message = client_socket.recv(1024).decode()
print("Client:", message)

num1, operator, num2 = message.split()
num1 = float(num1)
num2 = float(num2)

if operator == "+":
    result = num1 + num2
elif operator == "-":
    result = num1 - num2
elif operator == "*":
    result = num1 * num2
elif operator == "/":
    result = num1 / num2

reply = str(result)
client_socket.send(reply.encode())

client_socket.close()
server_socket.close()
```

client_math.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(("127.0.0.1", 8080))
print("Connected to server")

message = input("Enter operation (example: 5 + 3): ")
client_socket.send(message.encode())

reply = client_socket.recv(1024).decode()
print("Result:", reply)

client_socket.close()
```

Server Output (Math Operations)

\$ python3 server_math.py

```
Server is waiting for connection...
Client connected
Client: 5 + 3
$
```

Client Output (Math Operations)

```
$ python3 client_math.py
Connected to server
Enter operation (example: 5 + 3): 5 + 3
Result: 8.0
$
```